APDIO Workshop: Optimization Software & Practice

Part 2: Row Generation

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Roadmap

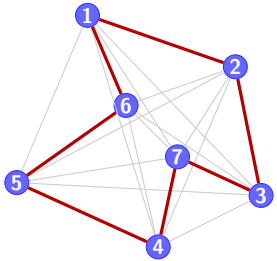
TSP → **Compact (MTZ)** → **Edge + Row Generation** → **Comparison**

- **Motivation:** some formulations have exponentially many constraints
- **Row generation:** add **constraints** dynamically as they are violated
- **SCIP plugin:** the constraint handler
- **Application:** Traveling Salesman Problem (TSP)

Traveling Salesman Problem (TSP)

Given: n cities, pairwise distances d_{ij}
Find: shortest tour visiting each city exactly once, returning to start

- One of the most famous NP-hard problems
- Symmetric TSP: $d_{ij} = d_{ji}$ (undirected edges)
- We will compare two formulations:
  1. Compact MTZ (polynomial constraints, weak LP)
  2. Edge formulation + SECs (exponential constraints, strong LP)

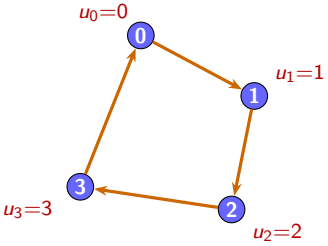


Optimal tour in red; complete graph in gray

MTZ Formulation

Variables: $x_{ij} \in \{0,1\}$ (directed edges), $u_i \in \mathbb{R}$ (positions)

$$
\begin{aligned}
\min \quad & \sum_{i \neq j} d_{ij}\, x_{ij} & & \\
\text{s.t.} \quad & \sum_{j \neq i} x_{ij} = 1 & \forall\, i \quad & \text{(leave once)} \\
& \sum_{i \neq j} x_{ij} = 1 & \forall\, j \quad & \text{(enter once)} \\
& u_i - u_j + n \cdot x_{ij} \leq n - 1 & \forall\, i, j \neq 0 \quad & \text{(MTZ)} \\
& x_{ij} \in \{0,1\} & &
\end{aligned}
$$



u_i values enforce a single directed tour

- $O(n^2)$ variables and constraints
- LP relaxation is **very weak**

Reference: Compact MTZ

Already Implemented

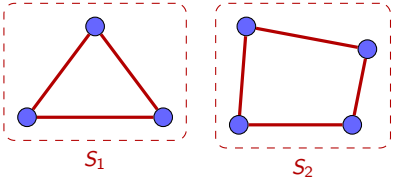
Study the MTZ formulation as a baseline before implementing row generation.

- File: `row_generation/compact_mtz.py`
- Run: `cd row_generation && python compact_mtz.py`

Edge Formulation with SECs

Variables: $x_e \in \{0,1\}$ for each undirected edge e

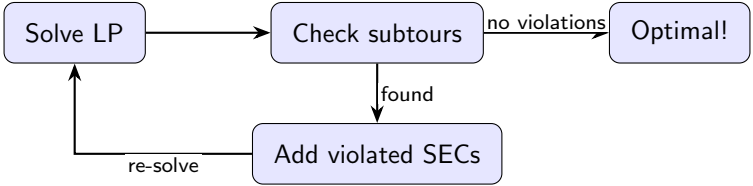
$$
\begin{aligned}
\min \quad & \sum_{e \in E} d_e \, x_e \\
\text{s.t.} \quad & \sum_{e \in \delta(v)} x_e = 2 && \forall\, v \in V \quad \text{(degree)} \\
& \sum_{e \in E(S)} x_e \leq |S| - 1 && \forall\, S \subset V \\
& 2 \leq |S| \leq n-1 && \text{(SECs)} \\
& x_e \in \{0,1\}
\end{aligned}
$$



Subtour violation!

- **Exponentially many** SECs: $O(2^n)$ subsets
- **Much stronger** LP relaxation than MTZ

Row Generation Paradigm



- Start with a relaxed model (only degree constraints)
- Iteratively find and add violated constraints
- Converges to optimality without enumerating all 2^n SECs

Constraint Handler in SCIP

SCIP implements row generation via the **constraint handler** plugin.

Callback	Purpose
`conscheck`	Check if a candidate integer solution is feasible
`consenfolp`	Enforce constraints on LP solution; add cuts if violated
`conslock`	Declare variable locks (required; can be empty)

```
class SubtourElimination(Conshdlr):
    def conscheck(self, constraints, solution, ...):
        # Find subtours; return FEASIBLE or INFEASIBLE
    def consenfolp(self, constraints, ...):
        # Add SEC for each subtour; return CONSADDED
```

Exercises: Subtour Detection & Constraint Handler

Exercise 1: Subtour Detection

Implement `find_subtours()` — find connected components in the edge graph.

- File: `row_generation/subtour.py`
- Test: `python row_generation/test_subtour.py`

Exercise 2: Constraint Handler

Implement `conscheck` and `consenfolp` callbacks.

- File: `row_generation/conshdlr_subtour.py`
- Test: `python row_generation/test_tsp.py`

Exercise: MTZ vs. Row Generation

Exercise 3: Experimental Comparison

Compare the compact MTZ formulation against row generation on instances of increasing size.

- File: `row_generation/experiment.py`
- Metrics to compare:
  - Solve time
  - Number of branch-and-bound nodes
  - LP relaxation gap
- Try instances with 10, 15, 20, 30 cities

What to expect

Row generation should dominate on larger instances thanks to its much stronger LP relaxation, despite the overhead of the separation callback.

Summary

What we covered:

- TSP: compact MTZ vs. edge formulation with SECs
- Row generation paradigm: add constraints on-the-fly
- SCIP constraint handler plugin (`conscheck`, `consenfolp`)
- Exponential constraint families handled implicitly

Key takeaway:

- A strong LP relaxation can make a huge difference in solving time
- Row generation lets us use formulations with exponentially many constraints

Part 3: Column generation & branch-and-price for bin packing